STPM Physics · Semester 1

DEFINITION LIST

Mechanics & Heat · Chapters 1–12. 171 definitions, laws and statements; key words in bold. Numbered chapter by chapter, taken from the PhysiQ notes.

CHAPTER 1 — PHYSICAL QUANTITIES

1.1 Physical Quantities & Units

1 **Base quantities** are physical quantities that are independent and cannot be defined in terms of other quantities. The seven SI base quantities and units are: length (metre, m), mass (kilogram, kg), time (second, s), electric current (ampere, A), thermodynamic temperature (kelvin, K), amount of substance (mole, mol) and luminous intensity (candela, cd).

2 **Derived quantities** are quantities defined in terms of base quantities by multiplication or division; their units are derived units (e.g. N = kg m s^{-2}, J = kg m^2 s^{-2}, W = kg m^2 s^{-3}, Pa = kg m^{-1} s^{-2}).

3 **Dimension** of a physical quantity is the way it is expressed in terms of the base quantities, written in square brackets, e.g. [L], [M], [T].

4 **Homogeneity principle**: every term in a physically correct equation must have the same dimensions (and units).

5 Prefixes: tera T 10^{12}, giga G 10^{9}, mega M 10^{6}, kilo k 10^{3}, deci d 10^{-1}, centi c 10^{-2}, milli m 10^{-3}, micro μ 10^{-6}, nano n 10^{-9}, pico p 10^{-12}.

6 Dimensionless quantities (pure numbers, angles in radian, ratios, refractive index, strain) have no unit.

1.2 Scalars and Vectors

7 **Scalar quantity** is a quantity that has magnitude only (e.g. distance, speed, mass, time, energy, temperature).

8 **Vector quantity** is a quantity that has both magnitude and direction (e.g. displacement, velocity, acceleration, force, momentum).

9 **Resultant** is the single vector that has the same effect as the vectors it replaces.

10 Vectors are added by the triangle law (head to tail) or the parallelogram law (tail to tail, resultant is the diagonal from the common tail).

11 **Vector subtraction**: A − B = A + (−B), where −B has the same magnitude as B but opposite direction.

12 **Resolution** of a vector is replacing it by two perpendicular components that together have the same effect.

13 **Unit vector** has magnitude 1 and gives direction only; a vector is written $A = A_x i + A_y j$.

1.3 Uncertainties in Measurements

14 **Accuracy** is the closeness of a measured value to the true (accepted) value; it is limited by systematic error.

15 **Precision** is the closeness of repeated measurements to one another (the spread of readings); it is limited by random error and indicated by a small uncertainty.

16 **Systematic error** is an error that makes all readings differ from the true value by the same amount in the same direction (e.g. zero error, wrongly calibrated instrument, parallax by a consistently wrong sight); it cannot be reduced by averaging, but can be corrected by checking zero or calibrating.

17 **Random error** is an error that causes readings to scatter above and below the true value in an unpredictable manner (e.g. reaction time, parallax by varying sight); it is reduced by taking many readings and calculating the mean.

18 **Absolute uncertainty** Δx is the uncertainty in a measurement, with the same unit; **fractional uncertainty** is $\frac{\Delta x}{x}$; **percentage uncertainty** is $\frac{\Delta x}{x} \times 100\%$.

19 Uncertainty of a single reading: analogue instrument ± half of the smallest division (some scales: ± one smallest division); digital instrument ± one smallest digit.

CHAPTER 2 — KINEMATICS

2.1 Linear Motion

1 **Distance** is the total length of the path travelled (scalar); **displacement** is the shortest straight-line distance from the initial to the final position in a stated direction (vector).

2 **Speed** is the rate of change of distance; **velocity** is the rate of change of displacement (vector).

3 **Instantaneous velocity** is the velocity at one instant, $v = \frac{ds}{dt}$, the gradient of the displacement–time graph at that instant.

4 **Acceleration** is the rate of change of velocity, $a = \frac{dv}{dt}$ (vector).

5 **Free fall** is motion under gravity alone, with constant downward acceleration $g = 9.81\ m\ s^{-2}$ near the Earth's surface (air resistance neglected), independent of mass.

6 A v–t graph: gradient = acceleration; area under graph = displacement (area below the t-axis is negative). An s–t graph: gradient = velocity. An a–t graph: area = change in velocity.

2.2 Projectiles

7 **Projectile** is an object projected into the air that moves under the action of gravity only (air resistance neglected).

8 The horizontal and vertical motions are **independent**: horizontally constant velocity ($a_x = 0$); vertically constant acceleration g downward.

9 The path (trajectory) of a projectile is a **parabola**.

10 At the highest point the vertical component of velocity is zero, the horizontal component is unchanged, and acceleration is g downward.

11 The speed at the same height on the way up and down is the same; the time of ascent equals the time of descent when landing at the launch level.

12 A projectile fired horizontally has zero initial vertical velocity; its time of fall depends only on the height h, not on the horizontal speed.

CHAPTER 3 — DYNAMICS

3.1 Newton's Laws of Motion

1 **Newton's first law**: a body remains at rest or moves with constant velocity in a straight line unless acted on by a resultant external force. **Inertia** is the property of a body that resists any change in its state of rest or uniform motion; it is measured by mass.

2 **Newton's second law**: the rate of change of momentum of a body is directly proportional to the resultant force acting on it and takes place in the direction of the force: $F = \frac{d(mv)}{dt}$.

3 **One newton** is the resultant force that gives a mass of 1 kg an acceleration of 1 m s^{-2} in the direction of the force.

4 **Newton's third law**: if body A exerts a force on body B, then B exerts an equal and opposite force on A. The pair acts on different bodies, is of the same type and acts along the same line at the same time.

5 **Weight** is the gravitational force on a body, W = mg, directed toward the centre of the Earth; **mass** is the amount of matter and is a scalar that does not change with place.

6 **Apparent weight** is the normal reaction exerted on a body, e.g. by a lift floor; it differs from the true weight when the body accelerates vertically.

3.2 Linear Momentum

7 **Linear momentum** of a body is the product of its mass and velocity, p = mv; a vector with the direction of the velocity, unit kg m s^{-1} or N s.

8 **Principle of conservation of linear momentum**: in an isolated system (no resultant external force), the total linear momentum before an interaction equals the total linear momentum after it.

9 **Impulse** is the product of the force and the time for which it acts, and equals the change in linear momentum of the body; it is a vector, unit N s.

10 Newton's second law (momentum form): the resultant force equals the rate of change of linear momentum.

11 A **force–time graph**: the area under the graph equals the impulse (change in momentum); the average force = impulse/time of contact.

12 Momentum is conserved in all collisions and explosions of an isolated system, whether elastic or inelastic.

3.3 Elastic and Inelastic Collisions

13 **Elastic collision** is a collision in which both total linear momentum and total kinetic energy are conserved.

14 **Inelastic collision** is a collision in which total linear momentum is conserved but total kinetic energy is not conserved (some is converted to heat, sound and deformation).

15 **Perfectly (completely) inelastic collision** is an inelastic collision in which the bodies stick together and move with a common velocity; the maximum loss of kinetic energy consistent with momentum conservation.

16 In an elastic collision in one dimension, the relative speed of approach equals the relative speed of separation.

17 In a two-dimensional collision, the total momentum is conserved separately along each of two perpendicular axes.

18 In an explosion, momentum is conserved and kinetic energy increases.

3.4 Friction

19 **Friction** is the force that opposes the relative motion, or the tendency of relative motion, between two surfaces in contact; it acts parallel to the surfaces.

20 **Static friction** acts when there is no relative motion; it equals the applied force up to a maximum value (limiting friction) $F_{max} = \mu_s N$.

21 **Kinetic (dynamic) friction** acts when the surfaces slide; it is approximately constant and independent of speed, $F = \mu_k N$, and usually less than the limiting static friction ($\mu_k < \mu_s$).

22 **Coefficient of friction** is the ratio of the friction force to the normal reaction, $\mu = \frac{F}{N}$; it has no unit and depends on the nature of the surfaces.

23 Laws of friction: friction is proportional to the normal reaction; it is independent of the area of contact (for the same normal reaction) and, for kinetic friction, independent of speed.

24 **Normal reaction** N is the perpendicular contact force exerted by a surface, not always equal to mg.

CHAPTER 4 — WORK, ENERGY AND POWER

4.1 Work Done by a Force

1 **Work done** by a constant force is the product of the force and the displacement in the direction of the force, $W = Fs\cos\theta$, where θ is the angle between the force and the displacement; a scalar, unit joule (J).

2 **One joule** is the work done when a force of 1 N moves its point of application through 1 m in the direction of the force.

3 The work done by a variable force is the area under the force–displacement graph, $W = \int F\,dx$.

4 Work is positive when the force has a component in the direction of displacement, zero when the force is perpendicular to the displacement (e.g. normal reaction, centripetal force), and negative when the force opposes the displacement (e.g. friction).

5 **Work–energy theorem**: the net work done on a body equals the change in its kinetic energy.

6 A force is **conservative** if the work done is independent of the path (e.g. gravity, spring force); friction is non-conservative.

4.2 Conservation of Energy

7 **Principle of conservation of energy**: energy cannot be created or destroyed; it can only be converted from one form to another, so the total energy of an isolated system is constant.

8 **Kinetic energy** is the energy a body possesses due to its motion, $E_k = \frac{1}{2}mv^2$.

9 **Gravitational potential energy** is the energy a body possesses due to its position in a gravitational field; near the Earth's surface the change in potential energy is $\Delta U = mg\Delta h$.

10 **Elastic potential energy** is the energy stored in a deformed spring, $U = \frac{1}{2}kx^2$.

11 **Conservation of mechanical energy**: if only conservative forces do work, the sum of kinetic and potential energy is constant. If friction or air resistance acts, mechanical energy decreases and is converted to heat.

12 A body released from rest from height h on a smooth surface reaches the bottom with speed $v = \sqrt{2gh}$, independent of the shape of the path.

4.3 Power and Efficiency

13 **Power** is the rate of doing work (or the rate of energy transfer), $P = \frac{W}{t}$; unit watt (W), $1\text{ W} = 1\text{ J s}^{-1}$.

14 **Instantaneous power** is the power at one instant, $P = \frac{dW}{dt} = Fv$, where v is the velocity in the direction of F.

15 **Average power** is the total work done divided by the total time taken.

16 **Efficiency** is the ratio of the useful output energy (or power) to the total input energy (or power), usually expressed as a percentage.

17 The efficiency of a real machine is less than 100% because some energy is lost as heat (friction, resistance) and sound; energy is still conserved.

18 **Kilowatt-hour** is a unit of energy: the energy transferred by a power of 1 kW in 1 hour.

CHAPTER 5 — CIRCULAR MOTION

5.1 Angular Velocity and Centripetal Acceleration

1 **Angular displacement** θ is the angle swept out by the radius vector from the centre of the circle, measured in radian. 1 rad is the angle subtended at the centre of a circle by an arc equal in length to the radius.

2 **Angular velocity** ω is the rate of change of angular displacement, $\omega = \frac{d\theta}{dt}$ (rad s^{-1}).

3 **Period** T is the time taken for one complete revolution; **frequency** f is the number of revolutions per unit time, $f = \frac{1}{T}$ (Hz).

4 **Uniform circular motion** is motion in a circle at constant speed. The velocity changes continuously in direction, so the object accelerates.

5 **Centripetal acceleration** is the acceleration of an object in circular motion; it is directed towards the centre of the circle, perpendicular to the velocity, and changes the direction of the velocity but not the speed.

6 Every point on a rigid rotating body has the same ω and T, but the linear speed v increases with distance r from the axis.

5.2 Centripetal Force

7 **Centripetal force** is the resultant force directed towards the centre of the circle that is required to make an object move in a circular path; it is perpendicular to the velocity and does no work.

8 Centripetal force is not a separate force; it is provided by tension, friction, gravitational force, normal reaction or a component of these forces.

9 **Newton's first law**: a body remains at rest or moves with constant velocity unless acted on by a resultant force; hence a body in circular motion needs a resultant force towards the centre.

10 Explain the "outward" sensation: it is due to inertia of the body, which tends to continue in a straight line (tangent); there is no real outward force on the body in an inertial frame.

11 If the centripetal force is removed (string breaks), the object moves along the tangent at constant velocity.

CHAPTER 6 — GRAVITATION

6.1 Newton's Law of Gravitation

1 **Newton's law of gravitation**: every point mass in the universe attracts every other point mass with a force that is directly proportional to the product of their masses and inversely proportional to the square of the distance between them.

2 The gravitational force is always attractive, acts along the line joining the two masses, and the two forces form an action–reaction pair (equal magnitude, opposite direction) on the two bodies.

3 A uniform spherical body acts as if its mass were concentrated at its centre, so r is measured between centres.

4 **Kepler's first law**: the orbit of a planet is an ellipse with the Sun at one focus.

5 **Kepler's second law**: the line joining a planet to the Sun sweeps out equal areas in equal times.

6 **Kepler's third law**: the square of the period of a planet is directly proportional to the cube of its mean orbital radius (semi-major axis), $T^2 \propto r^3$.

7 **Weight** is the gravitational force exerted on a body by the Earth; $g = \frac{GM}{R^2}$ at the surface.

6.2 Gravitational Field and Potential

8 **Gravitational field** is a region of space in which a mass experiences a gravitational force.

9 **Gravitational field strength** g at a point is the gravitational force per unit mass acting on a small test mass placed at that point; g is a vector directed towards the mass producing the field.

10 **Gravitational potential** V at a point is the work done per unit mass by an external force in bringing a small test mass from infinity to that point; V = 0 at infinity, so V is negative everywhere in the field of an attractive mass, and it is a scalar.

11 **Gravitational potential energy** U of a mass m at a point is the work done by an external force in bringing the mass from infinity to that point, U = mV.

12 Field strength is the negative of the potential gradient, $g = -\frac{dV}{dr}$; the area under a g–r graph gives $-\Delta V$.

13 **Escape speed** is the minimum speed at which a body must be projected from the surface of a planet so that it just reaches infinity, where its speed is zero.

14 An equipotential surface is a surface on which all points have the same potential; no work is done moving a mass along it, and g is perpendicular to it.

6.3 Satellite Motion

15 **Satellite** in a circular orbit is moving under the gravitational force of the central body, which provides the centripetal force: $\frac{GMm}{r^2} = \frac{mv^2}{r}$.

16 **Geostationary satellite** is a satellite that stays above a fixed point on the Earth's equator: it orbits in the equatorial plane, in the same direction as the Earth's rotation (west to east), with period 24 h (same angular velocity as the Earth), at a fixed height of about 3.6×10^7 m above the surface ($r \approx 4.2\times10^7$ m).

17 **Apparent weightlessness**: in orbit the satellite and its contents have the same acceleration towards the Earth (free fall), so there is no contact (normal) force between the astronaut and the floor, although the gravitational force still acts.

18 **Kepler's third law**: $T^2 \propto r^3$ for orbits about the same central mass.

19 The gravitational force on the satellite is perpendicular to its velocity in a circular orbit, so it does no work and the speed and total energy remain constant.

20 A satellite in a lower orbit has a greater speed and a smaller period; to move to a higher orbit the satellite gains total energy (E becomes less negative).

CHAPTER 7 — ROTATION OF RIGID BODIES

7.1 Torque and Angular Acceleration

1 **Torque (moment of a force)** about an axis is the product of the force and the perpendicular distance from the axis to the line of action of the force.

2 **Couple** is a pair of equal and opposite parallel forces whose lines of action do not coincide; it produces rotation without translation, and its torque is the product of one force and the perpendicular distance between the two forces.

3 **Newton's second law of rotation**: the angular acceleration of a rigid body about a fixed axis is directly proportional to the resultant torque and inversely proportional to its moment of inertia.

4 Torque is a vector (clockwise/anticlockwise sense must be stated); a force whose line of action passes through the axis has zero torque.

5 **Angular acceleration** α is the rate of change of angular velocity, $\alpha = \frac{d\omega}{dt}$ (rad s^{-2}).

7.2 Moment of Inertia

6 **Moment of inertia** I of a body about an axis is the sum of the products of the mass of each particle and the square of its perpendicular distance from the axis, $I = \Sigma mr^2$; it is the measure of the body's resistance to angular acceleration (rotational inertia).

7 Moment of inertia depends on the mass, on how the mass is distributed about the axis, and on the position of the axis; the same body has different I about different axes.

8 Moment of inertia is the rotational analogue of mass: $\tau = I\alpha$ corresponds to $F = ma$.

9 A body with its mass distributed further from the axis has a larger I and so is harder to start or stop rotating.

10 A flywheel has most of its mass at the rim to give a large I, so that it resists changes in angular speed (stores rotational kinetic energy).

7.3 Angular Momentum and Rotational Kinetic Energy

11 **Angular momentum** L of a rigid body about an axis is the product of its moment of inertia and its angular velocity, $L = I\omega$ (kg m^2 s^{-1}); for a point mass $L = mvr$.

12 **Principle of conservation of angular momentum**: the total angular momentum of a system remains constant if no resultant external torque acts on the system.

13 **Rotational kinetic energy** is the energy a rotating body possesses because of its angular velocity, $\frac{1}{2}I\omega^2$.

14 Newton's second law in terms of angular momentum: the resultant torque is the rate of change of angular momentum, $\tau = \frac{dL}{dt}$.

15 When the moment of inertia of a system decreases (e.g. a skater draws in the arms), ω increases so that L is constant; the rotational kinetic energy increases because work is done by the skater (internal forces).

16 In a collision or coupling of rotating bodies about the same axis (inelastic), angular momentum is conserved but kinetic energy is lost.

▶ CHAPTER 8 — STATICS

8.1 Equilibrium of Forces

1 **Translational equilibrium**: a body is in translational equilibrium when the resultant force acting on it is zero; it is at rest or moves with constant velocity.

2 **Rotational equilibrium**: a body is in rotational equilibrium when the resultant torque about any axis is zero.

3 **Conditions for equilibrium of a rigid body (static equilibrium)**: (1) the vector sum of all forces is zero, and (2) the sum of the moments about any point is zero.

4 **Triangle of forces**: if three coplanar forces acting at a point are in equilibrium, they can be represented in magnitude and direction by the three sides of a triangle taken in order.

5 For three non-parallel forces in equilibrium, the lines of action are coplanar and meet at one point (concurrent).

6 A couple is two equal and opposite parallel forces; a body acted on by a couple alone is in translational equilibrium but not in rotational equilibrium.

8.2 Moments and Centre of Gravity

7 **Moment of a force** about a point is the product of the force and the perpendicular distance from the point to the line of action of the force.

8 **Principle of moments**: for a body in rotational equilibrium, the sum of the clockwise moments about any point is equal to the sum of the anticlockwise moments about the same point.

9 **Centre of gravity** of a body is the point through which the whole weight of the body appears to act.

10 **Centre of mass** is the point at which the whole mass of the body can be considered to be concentrated; it coincides with the centre of gravity when the gravitational field is uniform.

11 The centre of gravity of a uniform regular body is at its geometric centre; for a freely suspended body it lies vertically below the point of suspension.

12 **Stable equilibrium**: a body returns to its original position after a small displacement; the centre of gravity rises on displacement. **Unstable**: the body moves further away; the centre of gravity falls. **Neutral**: the body stays in the new position; the centre of gravity stays at the same height.

13 A body topples when the vertical line through its centre of gravity falls outside its base.

CHAPTER 9 — DEFORMATION OF SOLIDS

9.1 Stress, Strain and Young's Modulus

1 **Stress** is the force acting normally per unit cross-sectional area of a material.

2 **Strain** is the ratio of the extension to the original length; it has no unit.

3 **Young modulus** is the ratio of tensile (or compressive) stress to the corresponding strain, within the limit of proportionality.

4 **Hooke's law**: the extension of a material is directly proportional to the applied force provided the limit of proportionality is not exceeded.

5 **Limit of proportionality** is the point on the graph beyond which stress is no longer proportional to strain. **Elastic limit** is the maximum stress beyond which the material does not return to its original length when the force is removed.

6 **Yield point** is the point at which the material extends considerably (plastic flow) with little or no increase in stress. **Ultimate tensile stress** is the maximum stress the material can withstand.

7 A **ductile** material shows a large plastic region before fracture (copper, mild steel); a **brittle** material fractures just after the elastic region with almost no plastic deformation (glass, cast iron); a **polymeric** material (rubber) shows a large non-linear extension and hysteresis.

9.2 Elastic Energy

8 **Elastic potential energy (strain energy)** is the energy stored in a material when it is deformed by a force within the elastic limit.

9 The work done in stretching a wire equals the area under the force–extension graph; this is stored as elastic potential energy if the elastic limit is not exceeded.

10 **Energy density** is the elastic potential energy stored per unit volume of the material.

11 Beyond the elastic limit, the work done is not wholly stored: part is dissipated as heat during plastic deformation, so the unloading line is not the loading curve (hysteresis for rubber: the area enclosed by the loop is the energy dissipated as heat).

CHAPTER 10 — GASES

10.1 Ideal Gas Equation

1 **Ideal gas** is a gas that obeys the equation of state pV = nRT at all pressures and temperatures.

2 **Boyle's law**: for a fixed mass of gas at constant temperature, the pressure is inversely proportional to the volume. **Charles's law**: at constant pressure, the volume is directly proportional to the thermodynamic temperature. **Pressure law**: at constant volume, the pressure is directly proportional to the thermodynamic temperature.

3 **Mole** is the amount of substance that contains as many elementary entities as there are atoms in 0.012 kg of carbon-12, i.e. the Avogadro constant N_A of entities.

4 A real gas behaves like an ideal gas at low pressure and high temperature (molecules far apart, intermolecular forces and molecular volume negligible).

5 **Dalton's law**: the total pressure of a mixture of gases equals the sum of the partial pressures of its components.

10.2 Kinetic Theory and Molecular Speeds

6 Assumptions of the kinetic theory of an ideal gas: (1) a gas consists of a very large number of identical molecules in continuous random motion; (2) the volume of the molecules is negligible compared with the volume of the container; (3) there are no intermolecular forces except during collisions; (4) collisions between molecules and with the walls are perfectly elastic; (5) the duration of a collision is negligible compared with the time between collisions; (6) Newton's laws of motion apply.

7 **Pressure** of a gas is caused by the molecules colliding with the walls: the rate of change of momentum of the molecules per unit area of the wall.

8 **Root-mean-square speed** c_{rms} is the square root of the mean of the squares of the speeds of the molecules.

9 The average translational kinetic energy of a molecule is directly proportional to the thermodynamic temperature, and is independent of the type of gas.

10 **Degrees of freedom** f is the number of independent ways in which a molecule can possess energy. Principle of equipartition of energy: each degree of freedom has an average energy of ½kT per molecule.

11 A monatomic gas has f = 3 (translational), a diatomic gas has f = 5 (3 translational + 2 rotational) at ordinary temperatures, and a polyatomic (non-linear) gas has f = 6.

CHAPTER 11 — THERMODYNAMICS

11.1 First Law of Thermodynamics

1 **First law of thermodynamics**: the increase in the internal energy of a system is equal to the heat supplied to the system minus the work done by the system on its surroundings (conservation of energy).

2 **Internal energy** of a system is the sum of the random kinetic energy and the potential energy of its molecules; for an ideal gas there is no intermolecular potential energy, so U depends only on the thermodynamic temperature.

3 **Heat** is the energy transferred between a system and its surroundings because of a temperature difference. **Work** is the energy transferred when the gas expands or is compressed against an external pressure.

4 **Molar heat capacity at constant volume** C_V is the heat required to raise the temperature of one mole of gas by 1 K at constant volume; **at constant pressure** C_p is the heat required for the same rise at constant pressure; C_p is greater than C_V because additional heat is needed for the work done in expansion.

11.2 Thermodynamic Processes

5 **Isothermal process** is a process at constant temperature: $\Delta U = 0$, so $Q = W$; heat absorbed is converted entirely into work done by the gas (pV = constant, Boyle's law).

6 **Adiabatic process** is a process in which no heat enters or leaves the system: $Q = 0$, so $\Delta U = -W$; an adiabatic expansion cools the gas and an adiabatic compression heats it.

7 **Constant-pressure (isobaric) process**: $W = p\Delta V$, and $\Delta U = Q - p\Delta V$. **Constant-volume (isochoric) process**: $W = 0$, so $Q = \Delta U$, and all the heat supplied increases the internal energy.

8 Work done by the gas equals the area under the p–V graph; in a cycle, the net work done equals the area enclosed by the loop (clockwise: net work done by the gas; anticlockwise: net work done on the gas).

9 An adiabatic curve is steeper than an isothermal curve through the same point on a p–V graph because γ is greater than 1.

10 **Reversible process** is one in which the system passes through a continuous series of equilibrium states and can be returned to the initial state with no change in the surroundings.

CHAPTER 12 — TRANSFER OF THERMAL ENERGY

12.1 Conduction, Convection and Radiation

1 **Conduction** is the transfer of heat through a material without bulk movement of the material, by lattice vibrations of atoms and, in metals, by the collisions of free electrons; it needs a medium.

2 **Convection** is the transfer of heat by the bulk movement of a fluid (liquid or gas) caused by density differences; **radiation** is the transfer of heat by electromagnetic waves (mainly infrared) and does not need a medium.

3 **Thermal conductivity** k is the rate of heat flow through a unit area of a material per unit temperature gradient, i.e. the heat transferred per unit time across unit area when the temperature gradient is 1 K m^{-1}.

4 **Stefan–Boltzmann law**: the total power radiated per unit surface area of a black body is directly proportional to the fourth power of its thermodynamic temperature.

5 A **black body** is a perfect absorber and a perfect emitter of radiation at all wavelengths (emissivity e = 1).

6 **Steady state**: the temperature at each point does not change with time, so the rate of heat flow is the same through every cross-section in series.